



Structured Floater Products Analysis

Case 2 A – Group 3
Financial Engineering

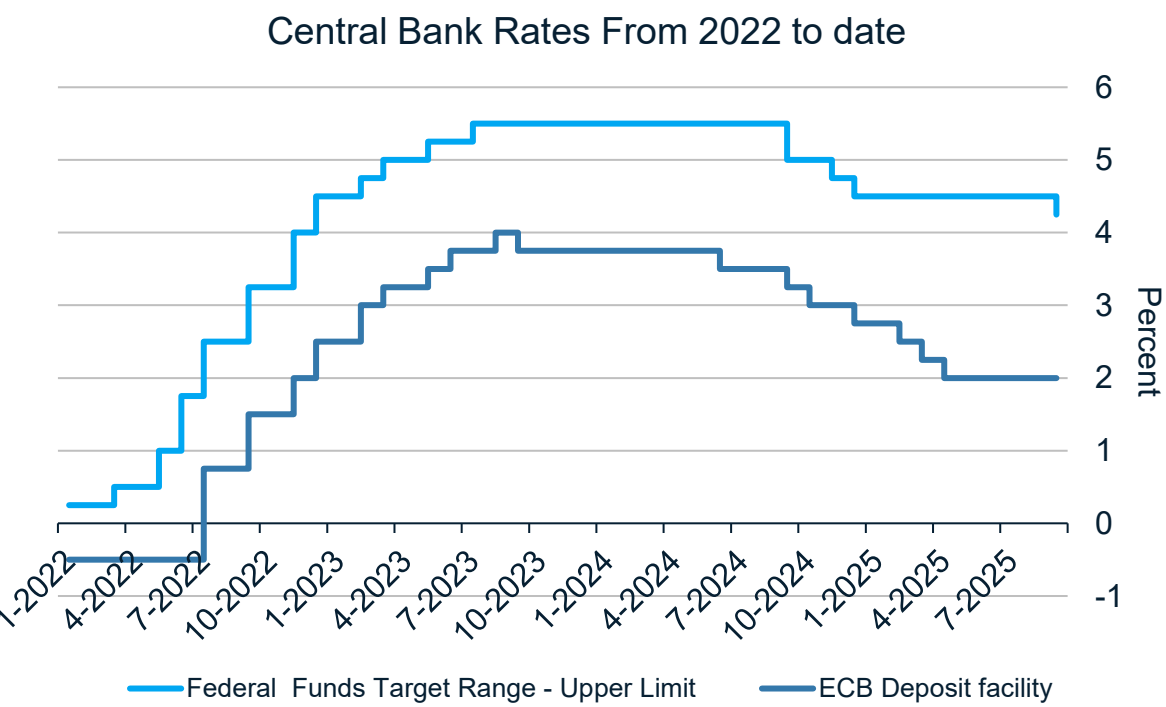
Presentation Structure

Our Goal through today's talk is to help you identify which structured floater provides optimal risk-adjusted returns, for whom, and under what market conditions?



- 1 Understanding Market Environment
- 2 Product Mechanics
- 3 Valuation Methodology
- 4 Interest Rate Scenarios
- 5 Risk-Return Analysis
- 6 Target Profiles and Use Cases
- 7 Key Value Proposition

Current Interest Rate Environment in EUR & USD



Market Pricing of the Central Bank Rates Movement

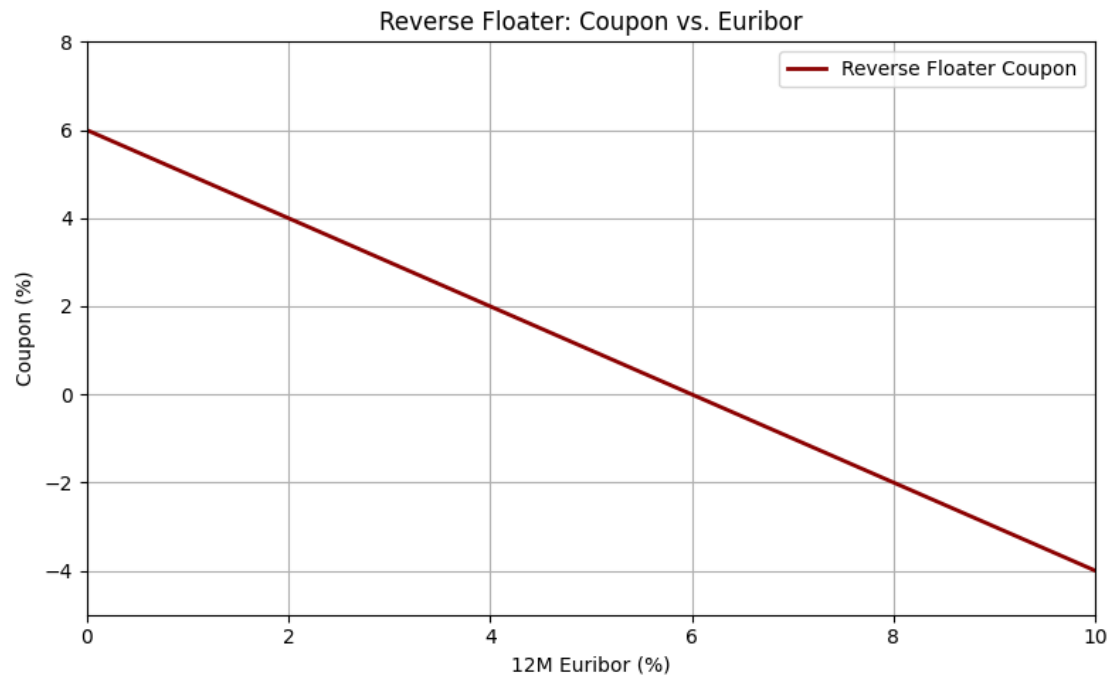
Probable Actions next FOMC Meeting		
Action	By	Probability %
Cut	-0.25	94.1
No Change	-	5.9

Probable Actions next ECB Governing Council		
Action	By	Probability %
No Change	-	98.4
Cut	-0.25	1.6

- EUR Rate: Large Volatility in the past years and stabilizing
- USD Rate: Elevated or Near Turning Point
- High Volatility Supports Derivatives & Structured Notes

Reverse Floater (X% – 12M Euribor) – Product Mechanics

A **Reverse Floater** is a structured bond whose coupon moves inversely to the reference rate – when rates rise, coupons fall, and vice versa



$X = 0.06$ (Fixed Rate = 6%), $m = 1.0$ (Coupon Leverage), $r_t = 12M$ Euribor

$$CPN_{Reverse Floater} = X - (m * r_t)$$

$CPN_{Reverse Floater}$: Floating Coupon of the Reverse Floater

X : Fixed Rate

m : Coupon Leverage

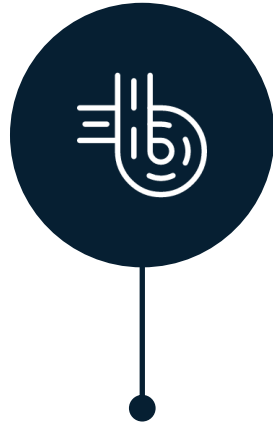
r_t : Reference Rate (e.g. 12M – EURIBOR)

Reverse Floater (X% – 12M Euribor) – Product Mechanics



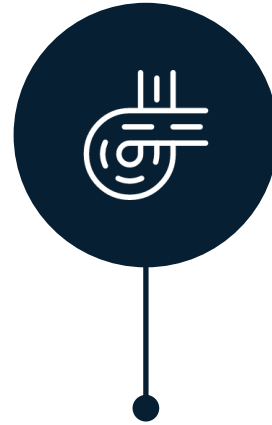
Inverse rate exposure

and high sensitivity to
interest rates



Yield Enhancement

if rates are expected to
decline



Implicit floor at 0%

provides downside
protection



Suitable for investors

with a declining-rate
view



Valuation Methodology – 5Y Reverse Floater (without volatility)

Replication Approach: Reverse Floater = Fixed Bond – Floating Note

$$P = \sum (X - L_t) \cdot P(0, t) + N \cdot P(0, T)$$

Fixed Bond ← $P = \boxed{\sum (X \cdot P(0, t)) + N \cdot P(0, T)} - \boxed{\sum (L_t \cdot P(0, t))}$ → Floating Note

Pricing Factors: The swap curve provides all the necessary inputs for valuation.



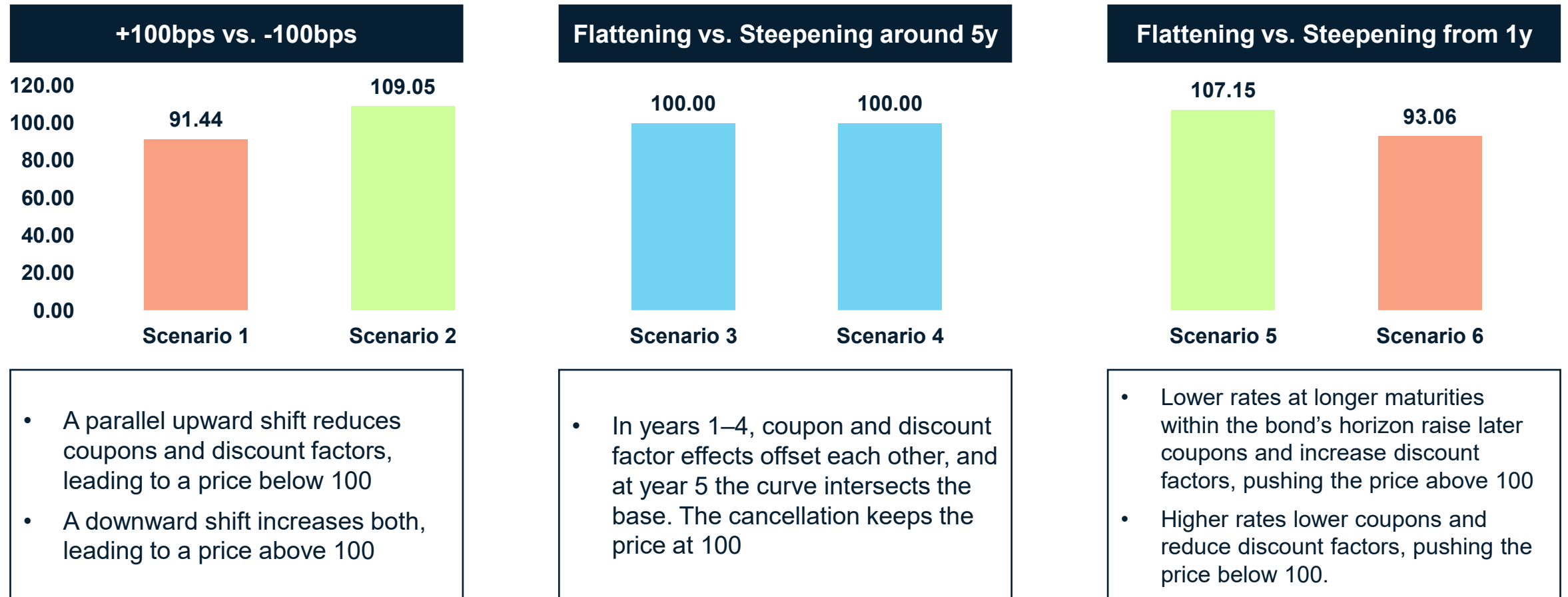
Single-curve DCF Approach

- **Future cash flow projection:** Use forward rates to approximate future Euribor for the floating leg; calculate the fixed leg directly by formula
- **Discounting:** Apply discount factors from the same curve to discount all cash flows

Par bond condition for X

Let Bond Price equals to 100 $\Rightarrow$ $X = 9.2\%$

Interest Rate Scenarios – 5Y Reverse Floater *(without volatility)*



“Parallel shifts have the strongest impact, intersection scenarios leave the price unchanged, and full-curve changes are dominated by long-term effects”

Valuation Methodology – 5Y Reverse Floater *(with volatility)*

Replication Approach: Reverse Floater = Fixed Bond - Floating Note – Cap (valued with Black-76, volatility = 20%)

Fixed Bond

$$\sum (X \cdot DF(t)) + N \cdot DF(T)$$

Floating Note

$$\sum (L_t \cdot DF(t))$$

Cap (Black-76)

$$\text{CapPV}(\sigma = 20\%)$$

Bond Price Formula

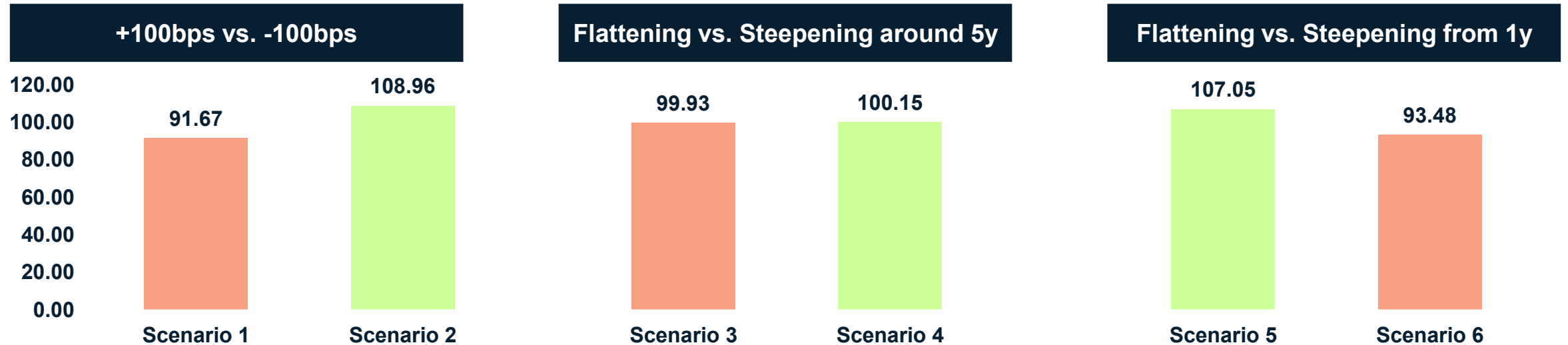
$$P = \sum (X - L_t) \cdot DF(t) + N \cdot DF(T) - \text{CapPV}(\sigma)$$

Where CapPV is computed with **Black-76**:

$$\text{CapletPV}_t = N \cdot DF(t) \cdot \Delta \cdot [F_t N(d_1) - K N(d_2)]$$

$$d_1 = \frac{\ln\left(\frac{F_t}{K}\right) + 0.5\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Interest Rate Scenarios – 5Y Reverse Floater *(with volatility)*



- The **0% coupon floor** introduces optionality in the reverse floater.
- This optionality can be valued as a **cap on the floating leg** (via Black-76 with $\sigma = 20\%$).
- Accounting for this option makes the bond slightly more valuable.
- To keep the price at par, the **fair fixed spread X is reduced: 9.20% → 9.1728%**.

Target Profile and Use Cases – Reverse Floater

A **Reverse Floater** is a bond + receiver swap + cap/floor → sold as a *Receiver note*.

Yield Seeking ← Stability	Investor Profile	Key Concern	Role of Floater	Typical Use Case
	Buy and Hold Investors	Focus on maturity value, less concerned with interim price swings	Provide coupon compensation, stabilize long-term income	Insurance companies, pension funds (small allocation)
	Income-Dependent Investors	Need steady cash flows to cover expenses	Higher coupons in falling-rate environments help offset lower reinvestment income	certain income-focused trusts or endowment funds
	Investors Exposed to Reinvestment Risk	Short-term instruments require frequent rollovers, earnings erode in low-rate environments	Coupons move inversely with rates, creating a natural hedge	Long-horizon investors seeking partial hedge against reinvestment risk
	Institutions with Risk Management Capacity	Can tolerate complexity, seek income enhancement; have tools to hedge	With derivatives or ALM tools, it can improve yield and stabilize cash flows when rates fall.	Large asset managers, sophisticated pension funds
	Opportunistic / Yield-Seeking Investors	Accept volatility and leverage for high-return potential; Declining rate view	Benefit from convex payoff in falling-rate scenarios	Hedge funds, family offices, high-net-worth individuals

Key Value Proposition

Reverse floaters provide income resilience, risk mitigation, and diversification benefits.

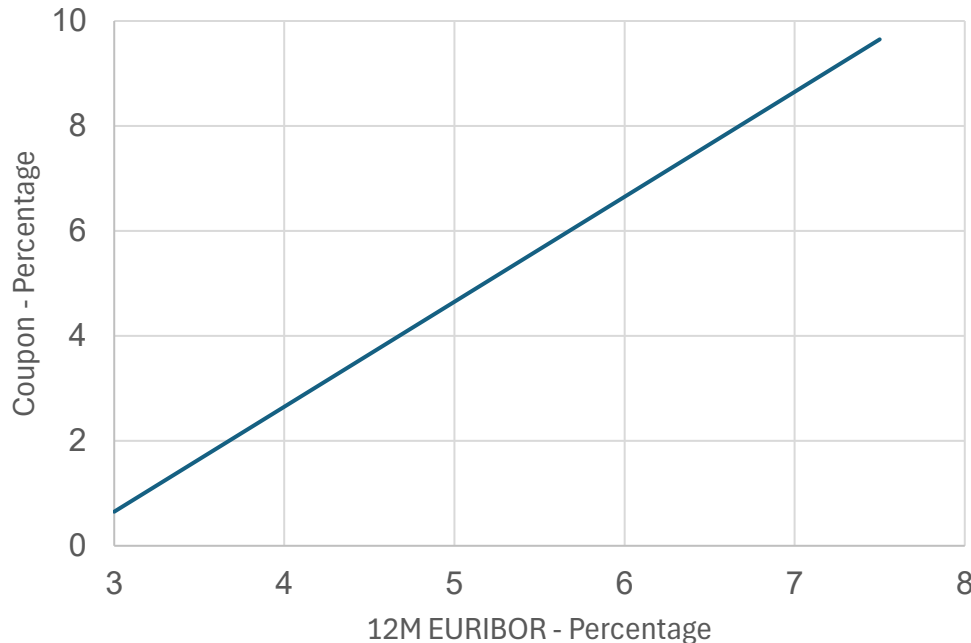
- ▶ **Enhanced Returns in Falling-Rate Environments** : Capture higher coupons when rates decline
- ▶ **Natural Hedge Against Reinvestment Risk**: Offsets income erosion from rolling over short-term instruments in a low-rate environment
- ▶ **Cash Flow Stabilization**: Provide predictable income streams to cover operating expenses or long-term liabilities, especially valuable for income-dependent investors
- ▶ **Convex Payoff / Yield Enhancement**: Offer non-linear upside in falling-rate scenarios; attractive for tactical positioning by opportunistic investors
- ▶ **Portfolio Diversification**: Add differentiated cash flow dynamics and yield enhancement to institutional portfolios, with low correlation to traditional bonds

Receiver with security ID number

Leveraged Floater (12M Euribor*2 – X) Product Mechanics

A **leveraged floater** is a floating-rate bond whose coupon is a leveraged multiple of a reference rate (e.g., $2 \times 12\text{M Euribor} - \text{spread}$), making its cash flows more sensitive to interest rate changes than a standard FRN

Leveraged Floater: Coupon vs. Euribor



$$CPN_{Leveraged Floater} = (m * r_t) - X$$

$CPN_{Leveraged Floater}$: Floating Coupon of the Leveraged Floater

X : Fixed Rate

m : Coupon Leverage

r_t : Reference Rate (e.g. 12M – EURIBOR)

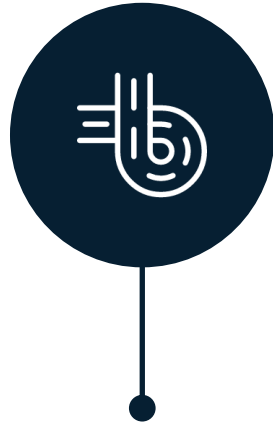
$X = 0.0535$ (Fixed Rate = 5.35%) $m = 2.0$ (Coupon Leverage), $r_t = 12\text{M Euribor}$

Leveraged Floater (12M Euribor*2 – X) Product Mechanics



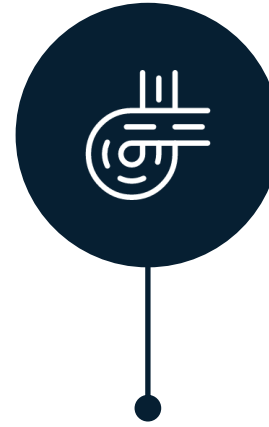
Leveraged rate exposure

and high sensitivity to
interest rates



Yield Enhancement

if rates are expected to
rise significantly



Often floor at 0%

to prevent negative
coupons



Suitable for investors

with a strong rising-rate
view



Valuation Methodology – 10Y Leveraged Floater (without volatility)

Replication Approach: Leverage Floater = 2 × Floating Note – Fixed Bond (+ Adjustment Zero-Coupon Bond)

$$P = \Sigma(2 \cdot L_t - X) \cdot P(0, t) + N \cdot P(0, T)$$
$$P = 2 \cdot \Sigma(L_t \cdot P(0, t)) - \Sigma(X \cdot P(0, t)) + N \cdot P(0, T)$$

Pricing Factors: The swap curve provides all the necessary inputs for valuation.



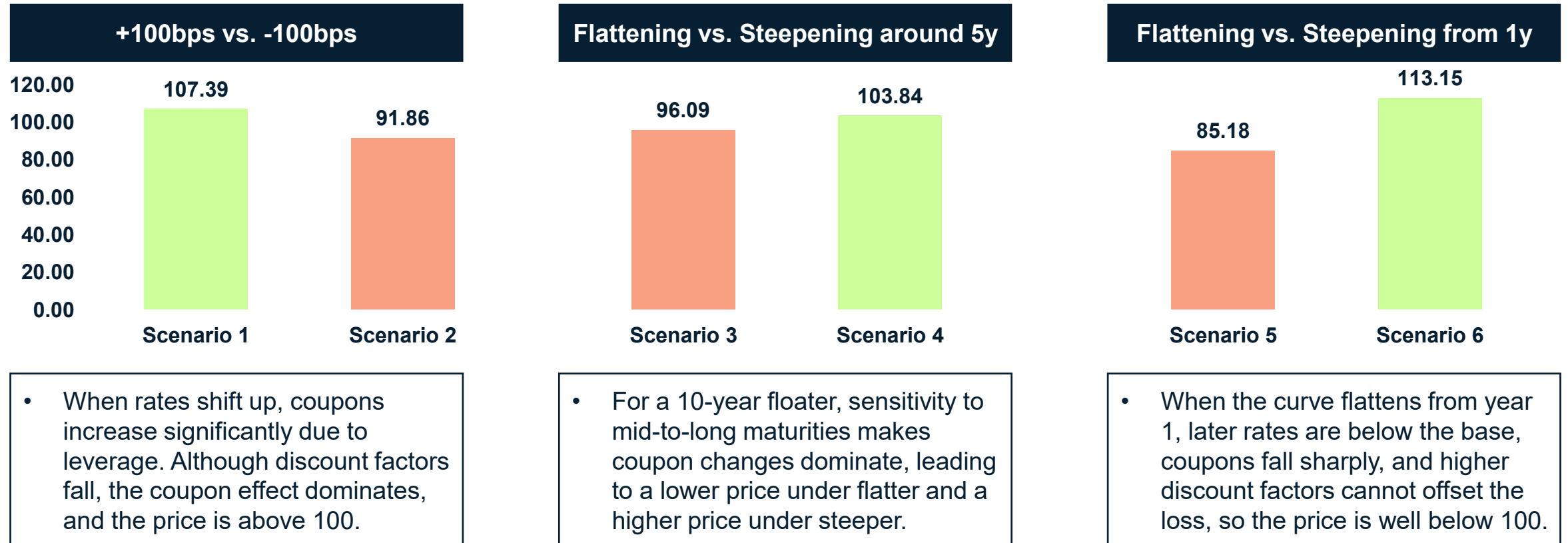
Single-curve DCF Approach

- **Future cash flow projection:** Use forward rates to approximate future Euribor for the floating leg; calculate the fixed leg directly by formula.
- **Discounting:** Apply discount factors from the same curve to discount all cash flows.

Par bond condition for X

Let Bond Price equal to 100 ➡ **$X = 5.35\%$**

Interest Rate Scenarios – 10Y Leveraged Floater *(without volatility)*



“For a 10-year leveraged floater, coupon sensitivity to rates is magnified. Full-curve changes have the largest price impact, as they alter coupon levels across the entire maturity horizon. Parallel shifts come next, while scenarios with intersection at year 5 have the smallest effect.”

Valuation Methodology – 10Y Leveraged Floater (volatility)

Replication Approach: Leverage Floater = 2 × Floating Note – Fixed Bond (+ Adjustment Zero-Coupon Bond)

$$P = \sum_{t=1}^{10} (2L_t - X)^+ \cdot DF_t \cdot N + N \cdot DF_{10}$$

Pricing Factors: The swap curve provides all the necessary inputs for valuation.



Single-curve DCF Approach

- **Future cash flow projection:** Use forward rates as underlying for Black-76 option valuation of the floored coupon.
- **Discounting:** Apply discount factors to the expected option-adjusted payoffs.

Par bond condition for X

Let Bond Price equal to 100 $\longrightarrow$ Solve for X with (with vol = 20%)

Valuation Methodology – 10Y Leveraged Floater *(with volatility)*

Replication Approach: Leveraged Floater = 2 x Floating Note - Fixed Bond + Coupon Floor (Black-76, vol=20%)

2 X Floating Note

$$PV_{\text{Floaters}} = 2 \cdot N \quad (\text{always par, } \approx 200)$$

- Fixed Bond (X)

$$PV_{\text{Fixed}} = X \cdot N \cdot \sum_{t=1}^{10} DF_t$$

+ Coupon Floor Option

$$PV_{\text{Floor}} = \sum_{t=1}^{10} 2 \cdot \text{Black-76 Call}(F_t, K = \frac{X}{2}, \sigma, T) \cdot DF_t \cdot N$$

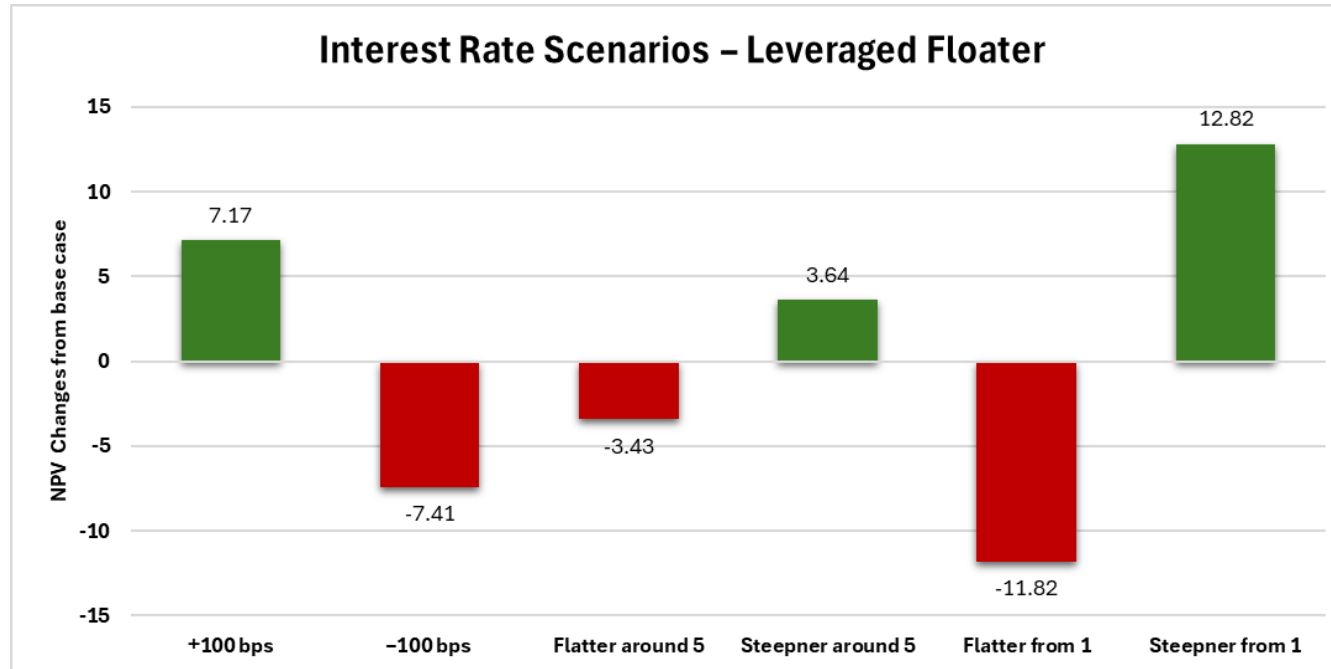
+ Redemption

$$PV_{\text{Redemption}} = N \cdot DF_{10}$$

= Final Equation

$$P = PV_{\text{Floaters}} - PV_{\text{Fixed}} + PV_{\text{Floor}} + PV_{\text{Redemption}}$$

Interest Rate Scenarios – Leveraged Floater *(with volatility)*



- A parallel upward shift increases long-term swap rates, raising the value of the payer swap and pushing the NPV above par.
- A downward shift lowers swap rates, reducing payer value, and drives NPV below par.
- Curve flattening reduces long-end rates relative to short-term Euribor, diminishing swap payoffs and pushing NPV below par.
- Curve steepening increases long-end rates relative to short-term Euribor, enhancing payer swap value and driving NPV above par.

"Parallel shifts drive the largest NPV swings, steepening boosts value while flattening erodes it, and long-term rates dominate overall performance due to leverage."

Target Profile and Use Cases – Leveraged Floater

A Leveraged Floater is effectively a bond + payer swap + options → sold to investors as a *Payer note*.

Stability ← Yield Seeking	Investor Profile	Key Concern	Role of Floater	Typical Use Case
	Rate Speculators	Expect rising interest rates, seek amplified exposure to rate movements and have high risk tolerance for volatility	Amplifies sensitivity to rate increases, Enhances coupon income in rising rate environments	Hedge funds, proprietary trading desks and sophisticated individual investors
	Duration Extension	Need to extend portfolio duration, Have limited capital for direct exposure and seeking efficient rate sensitivity	Achieves high effective duration with smaller capital and provides Cost-effective duration management	Asset managers with duration gaps, Banks managing duration gaps
	Focusing on Income Enhancement	Seeking higher yields in low-rate environments, Willing to accept rate risk for income and Need floating rate exposure	Higher coupons when rates rise, provides floating rate characteristics and Premium income vs. standard floaters	Income-focused mutual funds, High-yield bond portfolios and Credit investors seeking rate protection
	Tactical Allocation Trader	Short-term rate views, Portfolio rebalancing needs or Opportunistic positioning	Express directional rate bets efficiently, Quick adjustment of rate exposure and a Liquid alternative to futures/swaps	Multi-asset funds, Market timing portfolios

Key Value Proposition

Amplified Rate Exposure

- 2x-5x sensitivity to interest rate movements
- Direct speculation on rising rate environments
- Enhanced returns when directional views are correct

Portfolio Optimization Tool

- Duration extension without large capital commitment
- Gap management for institutional investors
- Liquidity alternative to derivatives (futures/swaps)

Income Enhancement Strategy

- Premium coupons in rising rate environments
- Floating rate protection against rate increases

Risk-Return Efficiency

- High effective duration with smaller notional investment
- Leveraged exposure without borrowing costs
- Clean rate exposure without credit risk overlay

Risk Analysis – Structured Floaters

Risk Sources	Direct Impact on Structured Floaters	Impact on Interest Rate Sensitivity
Swap Rate Level	Determines the fixed coupon component X at issuance (linked to swap of same maturity).	Larger $X \rightarrow$ scales up fixed-leg cashflows $\rightarrow$ indirectly raises modified duration.
Maturity (Tenor)	Defines the time horizon and average timing of cashflows.	Increases exposure to discounting changes $\rightarrow$ higher rate sensitivity
Coupon Leverage	Multiplies the floating-rate component of the coupon.	Higher leverage magnifies the bond's response to rate changes
Direction of the Floater	Determines whether coupon rises with rates (normal) or falls with rates (reverse).	Reverse floater amplifies duration risk; normal floater dampens rate sensitivity.
Convexity	Embedding of caps/floors introduces option-like features in cashflows	Option convexity becomes significant when coupon floors/caps bind (zero floor coupon)

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Thank you !